

KERACOLOR FF

High-performance, polymer-modified, water-repellent, cementitious grout with DropEffect® technology for joints up to 6 mm



BENEFITS AND FEATURES

- Water-repellent (DropEffect® Technology)
- Durable and strong
- Easy workability
- Easy to clean
- Uniform grouts, without cracks
- Good resistance to acids
- Very low emission of VOC

CLASSIFICATION IN COMPLIANCE WITH ISO 13007 AND EN 13888

Keracolor FF is an improved (2) cementitious (C) grout (G) with reduced water absorption (W) and high abrasion resistance (A) classified as CG2WA.

WHERE TO USE

Grouting interior and exterior floor and wall tiling (single-fired, double-fired, klinker, porcelain tile, etc.), stone material (natural stones, marble, granite, agglomerates, etc.), glass and marble mosaics.

Some application examples

- Grouting floor and wall ceramic tiles in interiors
- Grouting swimming pools
- Grouting particularly smooth and glossy floor surfaces (smooth and polished porcelain, pre-smoothed and polished marble, glazed tiles)
- Grouting ceramic tile industrial flooring where resistance to chemical agents is not required (garages, warehouses, etc.)

TECHNICAL CHARACTERISTICS

Keracolor FF is a blend of cement, very fine-graded aggregates, special polymers, additives and pigments. The use of special hydrophobic additives (DropEffect® technology) gives grouting mortars highly water-repellent properties, making them less prone to dirt pickup and with excellent durability.

The following features are obtained when mixed with the right water ratio and applied correctly:

- Water-repellent and droplet-effect
- Good compressive and flexural strength, good resistance to freeze/thaw cycles, and therefore good durability
- A smooth, easy-to-clean final surface, with low water absorption
- Very good abrasion resistance
- Low shrinkage, with the absence of cracks and fissures
- Good resistance to acids with pH > 3
- Excellent value for performance

When mixing **Keracolor FF** with **Fugolastic**, a special synthetic resin-based polymeric additive, the final characteristics are improved for use even under harsh conditions (grouting façades, swimming pools, bathrooms, floors with heavy traffic). For further information, consult the **Fugolastic** Technical Data Sheet.

RECOMMENDATIONS

- Do not mix **Keracolor FF** with cement or any other products. Never add water to the mixture that is beginning to set
- Never mix **Keracolor FF** with saline or dirty water
- Use the product in temperatures between +5°C and +40°C
- The amount of mixing water must be precisely measured. Excess water could induce the appearance of a whitish coat over the surface (efflorescence). Different colour tones could result if mixtures with different water ratios are prepared. Efflorescence over the surface is due to the formation of calcium carbonate and could also be caused by residual moisture contained in adhesives, not fully hydrated grouts, in substrates not adequately dried, or in substrates not adequately protected from rising damp
- After filling the joints, do not sprinkle **Keracolor** powder onto the grout, as this can cause uneven colour and reduce the mechanical strength
- Expansion joints and movement joints in floors and walls should never be filled with **Keracolor FF**, but with the appropriate flexible MAPEI sealant
- The surface of some ceramic tiles or stone materials is rough or contains micro-porosities. It is recommended to carry out a sample test to verify cleanability and, when necessary, apply a protective treatment over the surface, while avoiding penetration into the joints

APPLICATION PROCEDURE

Preparation of the joints

Before grouting, wait until the installation mortar or the adhesive has completely hardened. Verify that the waiting time indicated on the relevant technical data sheet has elapsed.

The joints must be clean, free of dust and raked back to at least 2/3 of the tile thickness. Excess adhesive or mortar should be removed whilst still fresh. Wet the joints with clean water when using very porous ceramic tiles in high temperatures and in windy conditions.

Preparation of the mix

While stirring, pour **Keracolor FF** into a clean, rust-free container containing 23-25% by weight, depending on grout colour, of clean, potable-grade water or **Fugolastic** (if required in the application).

Mix with a low-speed mixer to avoid air entrapment, producing a smooth consistency mix. Wait 2-3 minutes and briefly re-stir before use.

Use the mixture within 2 hours of preparation.

Application of the grout

Fill the joints completely with **Keracolor FF** using the appropriate MAPEI trowel or rubber float, making sure the joints are thoroughly compacted and free of unevenness. Remove excess **Keracolor FF**, while still fresh, from the

surface, moving the float diagonally across joints.

Finishing

When the mixture loses its plasticity and becomes matt, usually after 10-20 minutes, clean excess **Keracolor FF** with a damp hard cellulose sponge (e.g., MAPEI sponge) working diagonally to the joints. Rinse the sponge frequently using two separate buckets of water: one to remove the excess from the sponge and the other with clean water for rinsing out the sponge. Cleaning can also be carried out with a power float. To aid removal of the hardened product from the tiles, use a dampened Scotch-Brite® pad or a disc-type power float with special abrasive-felt discs. If cleaning is carried out too early (when the mixture is still plastic), the grout may be dragged from the joint, leading to colour variation. On the other hand, if the grout has already hardened, it is necessary to clean the surface mechanically, which could scratch the surface of the tiles.

When applying **Keracolor FF** in extremely hot, dry, or windy climates, it is recommended to wet the joints after several hours.

Wet curing of **Keracolor FF** always improves its final performance.

Any residual powdery haze can be cleaned from the surface with a clean, dry cloth. After the final cleaning, if the surface of the floors or walls are still contaminated with cementitious residues, an acid cleaner can be used (e.g. **UltraCare Keranet**), when the grout is completely cured.

If a product to remove grout residues at the time of application is required, the use of **UltraCare Keranet Easy**, suitable to remove excess grout from surfaces immediately after application, is recommended. For the use of the products from the **Ultracare** range, please refer to the relevant Technical Data Sheets.



Application of Keracolor FF with MAPEI float



Cleaning and finishing joints with a hard cellulose sponge



Floor grouted with Keracolor FF

SET TO LIGHT FOOT TRAFFIC

Floors are set to light foot traffic after approximately 24 hours.

READY FOR USE

(with hypothetical curing at +23°C and 50% R-H.)

Floors are ready for use after approximately 3 days. Swimming pools and tanks can be filled 7 days after grouting.

CLEANING

Clean tools and containers with plenty of water before **Keracolor FF** hardens.

CONSUMPTION

Consumption of **Keracolor FF** varies depending on the width of the joints, the size and thickness of the tiles. Please refer to the product calculator to estimate consumption rates at www.mapei.ae.

PACKAGING

20 kg paper bags and 4x5 kg alupacks.

COLOURS

Keracolor FF is available in 16 colours from the MAPEI range.

Keracolor FF		
100	WHITE	
103	MOON WHITE	
111	SILVER GREY	
112	MEDIUM GREY	
113	CEMENT GREY	
114	ANTHRACITE	
110	MANHATTAN 2000	
119	LONDON GREY	
130	JASMINE	
131	VANILLA	
137	CARIBBEAN	
132	BEIGE 2000	
141	CARMEL	
142	BROWN	
144	CHOCOLATE	
145	TERRA DI SIENA	

N.B.: Due to the printing processes involved, the colours should be taken as merely indicative of the shades of the actual product

STORAGE

When stored in dry conditions in the original, unopened bags, **Keracolor FF** has a shelf life of 12 months for 20 kg and 24 months for 4x5 kg alupack boxes. Always refer to what is written on the packaging.

CONSUMPTION TABLE DEPENDING ON THE SIZE OF THE TILE AND WIDTH OF THE JOINTS (kg/m²)

Size of the tile (mm)	Width of the joint (mm)		
	2	3	5
75x150x6	0.4	0.5	0.9
100x100x7	0.4	0.6	1.1
100x100x9	0.5	0.8	1.4
150x150x6	0.2	0.4	0.6
200x200x7	0.2	0.3	0.5
200x200x9	0.3	0.4	0.7
300x300x10		0.3	0.5
300x300x20		0.6	1.0
300x600x10		0.2	0.4
400x400x10		0.2	0.4
500x500x10			0.3
600x600x10			0.3
750x750x10			0.2
100x600x9			0.8
150x600x9			0.6
150x900x9			0.5
150x1200x10			0.6
225x450x9			0.5
225x900x9			0.4
250x900x9			0.3
250x1200x10			0.4
600x600x5			0.1
600x600x3			0.1

CONSUMPTION CALCULATION FORMULA:

$$\frac{(A + B)}{(A \times B)} \times C \times D \times 1.5 = \frac{\text{kg}}{\text{m}^2}$$

A = length of tile (in mm)

B = width of the tile (in mm)

C = thickness of the tile (in mm)

D = width of the joint (in mm)

TECHNICAL DATA (typical values)

In compliance with:

- European EN 13888-1 as CG2WA
- ISO 13007-3 as CG2WA
- American ANSI A118.6

PRODUCT IDENTIFICATION

Type:

fine powder

Colour:

16 colours from the MAPEI range

Bulk density:

1000-1400 kg/m³

Dry solids content:

100%

EMICODE:

EC1 Plus - very low emission

APPLICATION DATA (at +23°C and 50% R.H.)

Mixing ratio:	100 parts of Keracolor FF with 23-25 parts of water by weight depending on the colour
Consistency of mix:	fluid paste
Density of the mix:	2000 kg/m ³
pH of mix:	approx. 13
Pot life of mix:	approx. 2 hours
Application temperature:	from +5°C to +40°C
Grouting after installation: – on walls bonded with normal setting adhesive: – on walls bonded with fast setting adhesive: – on walls laid with mortar: – on floors bonded with normal setting adhesive: – on floors bonded with fast setting adhesive: – on floors laid with mortar:	4-8 hours 1-2 hours 2-3 days 24 hours 3-4 hours 7-10 days
Waiting time for finishing:	10-20 minutes
Set to light foot traffic:	24 hours
Ready for use:	3 days (7 days for tanks and swimming pools)

FINAL PERFORMANCE

Flexural strength after 28 days (EN 13888-2): Compressive strength after 28 days (EN 13888-2): Flexural strength after freeze-thaw cycles (EN 13888-2): Compressive strength after freeze-thaw cycles (EN 13888-2): Abrasion resistance (EN 13888-2): Shrinkage (EN 13888-2): Water absorption after 30 min. (EN 13888-2): Water absorption after 4 hours (EN 13888-2):	In compliance with ISO 13007-3 as CG2WA and with European Norm EN 13888-1 as CG2WA
Resistance to moisture:	excellent
Resistance to ageing:	excellent
Resistance to solvents, oils and alkalis:	excellent
Resistance to acids:	good if pH > 3
Resistance to temperature:	from -30°C to +80°C
VOC (SCAQMD 1168):	0 g/l

SAFETY INSTRUCTIONS FOR THE PREPARATION AND INSTALLATION

Instructions for the safe use of our products can be found on the latest version of the Safety Data Sheet, available from our website www.mapei.ae.

IMPORTANT NOTES

While we try to ensure that any advice, recommendations, or information in our literature is accurate and correct, we have no control over the circumstances in which our product is used. It is, therefore, important that installers satisfy themselves that the product and conditions are suitable for the envisaged application. No warranty can be given, nor can responsibility be accepted other than the product we supply will meet our written specifications. The installer should ensure that our latest product data and safety information sheets have been consulted before use.

Please refer to the current version of the Technical Data Sheet, available from our website www.mapei.ae.

LEGAL NOTICE

The contents of this Technical Data Sheet ("TDS") may be copied into other project-related documents. The resulting document shall not supplement or replace requirements per the TDS in force at the time of the MAPEI product installation. The most up-to-date TDS can be downloaded from our website, www.mapei.ae.

ANY ALTERATIONS TO THE WORDING OR REQUIREMENTS CONTAINED OR DERIVED FROM THIS TDS EXCLUDE THE RESPONSIBILITY OF MAPEI.

All relevant references for the product are available upon request and from www.mapei.com.

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